

2014 January Q6 C34

6. Given that the binomial expansion, in ascending powers of x , of

$$\frac{6}{\sqrt{9 + Ax^2}}, \quad |x| < \frac{3}{\sqrt{|A|}}$$

is $B - \frac{2}{3}x^2 + Cx^4 + \dots$

- (a) find the values of the constants A , B and C .

(7)

- (b) Hence find the coefficient of x^6

(2)

2014 June Q5 C34

5.

$$f(x) = (8 + 27x^3)^{\frac{1}{3}}, \quad |x| < \frac{2}{3}$$

Find the first three non-zero terms of the binomial expansion of $f(x)$ in ascending powers of x . Give each coefficient as a simplified fraction.

(5)

2015 January Q5 C34

5. (a) Use the binomial expansion, in ascending powers of x , of $\frac{1}{\sqrt{1-2x}}$ to show that

$$\frac{2+3x}{\sqrt{1-2x}} \approx 2 + 5x + 6x^2, \quad |x| < 0.5$$

(4)

- (b) Substitute $x = \frac{1}{20}$ into

$$\frac{2+3x}{\sqrt{1-2x}} = 2 + 5x + 6x^2$$

to obtain an approximation to $\sqrt{10}$

Give your answer as a fraction in its simplest form.

(3)

2015 June Q2 C34

2. Given that

$$\frac{4(x^2 + 6)}{(1 - 2x)(2 + x)^2} \equiv \frac{A}{(1 - 2x)} + \frac{B}{(2 + x)} + \frac{C}{(2 + x)^2}$$

(a) find the values of the constants A and C and show that $B = 0$

(4)

(b) Hence, or otherwise, find the series expansion of

$$\frac{4(x^2 + 6)}{(1 - 2x)(2 + x)^2}, \quad |x| < \frac{1}{2}$$

in ascending powers of x , up to and including the term in x^2 , simplifying each term.

(5)

2016 January Q1 C34

1.
$$f(x) = (3 - 2x)^{-4}, \quad |x| < \frac{3}{2}$$

Find the binomial expansion of $f(x)$, in ascending powers of x , up to and including the term in x^2 , giving each coefficient as a simplified fraction.

(4)

2016 June Q3 C34

3. (a) Find the binomial expansion of

$$(1 + ax)^{-3}, \quad |ax| < 1$$

in ascending powers of x , up to and including the term in x^3 , giving each coefficient as simply as possible in terms of the constant a .

(3)

$$f(x) = \frac{2 + 3x}{(1 + ax)^3}, \quad |ax| < 1$$

In the series expansion of $f(x)$, the coefficient of x^2 is 3

Given that $a < 0$

- (b) find the value of the constant a ,

(4)

- (c) find the coefficient of x^3 in the series expansion of $f(x)$, giving your answer as a simplified fraction.

(2)

2017 January Q3 C34

3. (a) Express $\frac{9 + 11x}{(1 - x)(3 + 2x)}$ in partial fractions. (3)

- (b) Hence, or otherwise, find the series expansion of

$$\frac{9 + 11x}{(1 - x)(3 + 2x)}, \quad |x| < 1$$

in ascending powers of x , up to and including the term in x^3 .
Give each coefficient as a simplified fraction.

(6)

2017 June Q4 C34

4.

$$f(x) = \frac{27}{(3 - 5x)^2} \quad |x| < \frac{3}{5}$$

- (a) Find the series expansion of $f(x)$, in ascending powers of x , up to and including the term in x^3 . Give each coefficient in its simplest form.

(5)

Use your answer to part (a) to find the series expansion in ascending powers of x , up to and including the term in x^3 , of

(b) $g(x) = \frac{27}{(3 + 5x)^2} \quad |x| < \frac{3}{5}$

(1)

(c) $h(x) = \frac{27}{(3 - x)^2} \quad |x| < 3$

(2)

2017 October Q7 C34

7. (a) Use the binomial series to expand

$$\frac{1}{(2-3x)^3} \quad |x| < \frac{2}{3}$$

in ascending powers of x , up to and including the term in x^2 , giving each term as a simplified fraction.

(5)

$$f(x) = \frac{4+kx}{(2-3x)^3} \quad \text{where } k \text{ is a constant and } |x| < \frac{2}{3}$$

Given that the series expansion of $f(x)$, in ascending powers of x , is

$$\frac{1}{2} + Ax + \frac{81}{16}x^2 + \dots$$

where A is a constant,

- (b) find the value of k ,

(2)

- (c) find the value of A .

(2)

2018 January Q2 C34

2.

$$f(x) = (125 - 5x)^{\frac{2}{3}} \quad |x| < 25$$

- (a) Find the binomial expansion of $f(x)$, in ascending powers of x , up to and including the term in x^2 , giving the coefficient of x and the coefficient of x^2 as simplified fractions. **(4)**
- (b) Use your expansion to find an approximate value for $120^{\frac{2}{3}}$, stating the value of x which you have used and showing your working. Give your answer to 5 decimal places. **(3)**

2018 June Q4 C34

4. (a) Find the binomial expansion of

$$(1 + px)^{-4}, \quad |px| < 1$$

in ascending powers of x , up to and including the term in x^3 , giving each coefficient as simply as possible in terms of the constant p .

(3)

$$f(x) = \frac{3 + 4x}{(1 + px)^4} \quad |px| < 1$$

where p is a positive constant.

In the series expansion of $f(x)$, the coefficient of x^2 is twice the coefficient of x .

- (b) Find the value of p .

(5)

- (c) Hence find the coefficient of x^3 in the series expansion of $f(x)$, giving your answer as a simplified fraction.

(2)

2018 October Q6 C34

6. (a) Use binomial expansions to show that, for $|x| < \frac{1}{2}$

$$\sqrt{\frac{1+2x}{1-x}} \approx 1 + \frac{3}{2}x + \frac{3}{8}x^2 \quad (6)$$

- (b) Find the exact value of $\sqrt{\frac{1+2x}{1-x}}$ when $x = \frac{1}{10}$

Give your answer in the form $k\sqrt{3}$, where k is a constant to be determined.

(1)

- (c) Substitute $x = \frac{1}{10}$ into the expansion given in part (a) and hence find an approximate value for $\sqrt{3}$

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

(2)

2019 January Q2 C34

2. Given that

$$\frac{3x^2 + 4x - 7}{(x+1)(x-3)} \equiv A + \frac{B}{x+1} + \frac{C}{x-3}$$

(a) find the values of the constants A , B and C .

(4)

(b) Hence, or otherwise, find the series expansion of

$$\frac{3x^2 + 4x - 7}{(x+1)(x-3)} \quad |x| < 1$$

in ascending powers of x , up to and including the term in x^2

Give each coefficient as a simplified fraction.

(6)

2019 June Q10 C34

10. (a) Use the binomial series to find the expansion of

$$\frac{1}{(2+3x)^3} \quad |x| < \frac{2}{3}$$

in ascending powers of x , up to and including the term in x^2 , giving each term as a simplified fraction.

(5)

- (b) Hence or otherwise, find the coefficient of x^2 in the series expansion of

(i) $\frac{1}{(2+6x)^3} \quad |x| < \frac{1}{3}$

(ii) $\frac{4-x}{(2+3x)^3} \quad |x| < \frac{2}{3}$

(4)

2020 October Q2 P4

2. (a) Use the binomial expansion to expand

$$(4 - 5x)^{-\frac{1}{2}} \quad |x| < \frac{4}{5}$$

in ascending powers of x , up to and including the term in x^2 giving each coefficient as a fully simplified fraction.

(4)

$$f(x) = \frac{2 + kx}{\sqrt{4 - 5x}} \quad \text{where } k \text{ is a constant and } |x| < \frac{4}{5}$$

Given that the series expansion of $f(x)$, in ascending powers of x , is

$$1 + \frac{3}{10}x + mx^2 + \dots \quad \text{where } m \text{ is a constant}$$

- (b) find the value of k ,

(2)

- (c) find the value of m .

(2)

2021 January Q1 P4

1. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(\frac{1}{4} - 5x\right)^{\frac{1}{2}} \quad |x| < \frac{1}{20}$$

giving each coefficient in its simplest form.

(5)

By substituting $x = \frac{1}{100}$ into the answer for (a),

- (b) find an approximation for $\sqrt{5}$

Give your answer in the form $\frac{a}{b}$ where a and b are integers to be found.

(2)

2021 June Q1 P4

1. Given that k is a constant and the binomial expansion of

$$\sqrt{1+kx} \quad |kx| < 1$$

in ascending powers of x up to the term in x^3 is

$$1 + \frac{1}{8}x + Ax^2 + Bx^3$$

- (a) (i) find the value of k ,
(ii) find the value of the constant A and the constant B .

(5)

- (b) Use the expansion to find an approximate value to $\sqrt{1.15}$

Show your working and give your answer to 6 decimal places.

(2)

2021 October Q4 P4

4.

$$f(x) = \sqrt{1 - 4x^2} \quad |x| < \frac{1}{2}$$

- (a) Find, in ascending powers of x , the first four non-zero terms of the binomial expansion of $f(x)$. Give each coefficient in simplest form.

(4)

- (b) By substituting $x = \frac{1}{4}$ into the binomial expansion of $f(x)$, obtain an approximation for $\sqrt{3}$

Give your answer to 4 decimal places.

(2)

2022 January Q2 P4

2. (a) Find, in ascending powers of x , the first three non-zero terms of the binomial series expansion of

$$\sqrt[3]{1+4x^3} \quad |x| < \frac{1}{\sqrt[3]{4}}$$

giving each coefficient as a simplified fraction.

(4)

- (b) Use the expansion from part (a) with $x = \frac{1}{3}$ to find a rational approximation to $\sqrt[3]{31}$

(3)

2022 June Q1 P4

1. The binomial expansion of

$$(3 + kx)^{-2} \quad |kx| < 3$$

where k is a non-zero constant, may be written in the form

$$A + Bx + Cx^2 + Dx^3 + \dots$$

where A , B , C and D are constants.

(a) Find the value of A

(1)

Given that $C = 3B$

(b) show that

$$k^2 + 6k = 0$$

(3)

(c) Hence (i) find the value of k

(ii) find the value of D

(3)

2022 October Q4 P4

4.

$$g(x) = \frac{1}{\sqrt{4-x^2}}$$

- (a) Find, in ascending powers of x , the first four non-zero terms of the binomial expansion of $g(x)$. Give each coefficient in simplest form.

(5)

- (b) State the range of values of x for which this expansion is valid.

(1)

- (c) Use the expansion from part (a) to find a fully simplified rational approximation for $\sqrt{3}$

Show your working and make your method clear.

(2)

2023 January Q1 P4

1.

$$f(x) = \frac{5x + 10}{(1 - x)(2 + 3x)}$$

(a) Write $f(x)$ in partial fraction form.

(3)

(b) (i) Hence find, in ascending powers of x up to and including the terms in x^2 , the binomial series expansion of $f(x)$. Give each coefficient as a simplified fraction.

(5)

(ii) Find the range of values of x for which this expansion is valid.

(1)

2023 June Q1 P4

1. (a) Find the first 4 terms of the binomial expansion, in ascending powers of x , of

$$\left(\frac{1}{4} - \frac{1}{2}x\right)^{-\frac{3}{2}} \quad |x| < \frac{1}{2}$$

giving each term in simplest form.

(5)

Given that

$$\left(\frac{1}{4} - \frac{1}{2}x\right)^n \left(\frac{1}{4} - \frac{1}{2}x\right)^{-\frac{3}{2}} = \left(\frac{1}{4} - \frac{1}{2}x\right)^{\frac{1}{2}}$$

- (b) write down the value of n .

(1)

- (c) Hence, or otherwise, find the first 3 terms of the binomial expansion, in ascending powers of x , of

$$\left(\frac{1}{4} - \frac{1}{2}x\right)^{\frac{1}{2}} \quad |x| < \frac{1}{2}$$

giving each term in simplest form.

(3)

2023 October Q1 P4

1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\frac{8}{(2 - 5x)^2}$$

writing each term in simplest form.

(4)

- (b) Find the range of values of x for which this expansion is valid.

(1)

2024 January Q1 P4

1. Find, in ascending powers of x up to and including the term in x^3 , the binomial expansion of

$$(1 - 4x)^{-3} \qquad |x| < \frac{1}{4}$$

fully simplifying each term.

(4)